

BUILD OPS — Prompt Pack

Copy-paste prompts for executing the refactor

Companion to `docs/PRD.md` v1.3 · 12 August 2026 · Toolchain: OpenAI Codex CLI

How to use this pack

1. Commit `AGENTS.md` at the repo root and this PRD as `docs/PRD.md`. Codex loads `AGENTS.md` automatically at session start; the PRD is opened on demand.
2. Launch Codex **from the repo root** (`codex`), not from a subdirectory — the instruction walk starts at the git root.
3. Start every session with **Prompt A (Kickoff)**.
4. Then paste exactly **one** work-order prompt (B-series).
5. Use the C-series prompts when something goes wrong or needs review.
6. **One work order per session.** When a work order is done, quit and start a fresh one. Long sessions drift, and drift in a schema refactor is expensive.

Note on `AGENTS.md` : Codex builds its instruction chain once, at session start. If you edit `AGENTS.md`, restart the session or the change has no effect. And remember it is instruction, not enforcement — for `DROP` statements and production pushes, rely on sandbox permissions and approval settings, not on the file asking nicely.

Text in `[SQUARE BRACKETS]` is for you to fill in.

A · Session kickoff — paste at the start of every session

You are a senior engineer working on BUILD OPS, a construction ERP for Actuate Builders Inc. (ABI), a Philippine design-and-build, fit-out and MEP contractor. The codebase is a Next.js app on Vercel with Supabase/Postgres.

Read `docs/PRD.md` in full before writing any code or SQL.

This is a REFACTOR, not a rewrite. Three schema facts were verified against the live database and are not open for discussion:

1. Money is BIGINT centavos; percentages are integer basis points. No floats.
2. tenant_id is NOT NULL on all application tables, with RLS already in place.
3. bom_line_items.id is a stable UUID already referenced by cost, budget, PO and RFQ records.

Therefore the relational spine already exists. The defect is GRAIN and PRICING, not identity. Do not propose rebuilding what already works.

Hard constraints for this session:

- Additive migrations only. No DROP COLUMN, no DROP TABLE, no re-pointing an existing foreign key.
- Do NOT create a scope_items table or a scope_item_id column. bom_line_items IS the scope item. See ADR-03.
- Do NOT build, extend or touch the general ledger, journals, posting periods, or reconciliation. ABI runs SAP. See ADR-07.
- No float, double precision, unscaled numeric, or JS number in any monetary path.
- Every new table carries tenant_id NOT NULL with the matching RLS policy.
- Every acceptance criterion ships as an automated test, not a manual check.

Before you write anything, confirm back to me in under 150 words:

- (a) which work order we are doing,
- (b) the files and tables you expect to touch,
- (c) anything in the PRD you think is wrong or ambiguous.

Then stop and wait for my go-ahead.

B • Work-order prompts

Paste one of these after Prompt A. Each assumes the PRD is in context.

B-00 • CI gate

Execute WO-00 from docs/PRD.md.

Set up CI to run the full test suite on every PR, plus three static checks that must fail the build:

1. Any migration introducing float, double precision, or unscaled numeric on a column whose name matches /(cost|price|rate|amount|total|centavos|value)/
2. Any CREATE TABLE without tenant_id NOT NULL
3. Any record with an E2E_ prefix in a non-demo tenant

Write a deliberately-broken fixture for each of the three so I can see the build fail, then remove the fixtures.

B-01 • Purge demo data

Execute WO-01 from docs/PRD.md.

The live build shows these contaminated records:

- E2E_QA_20260513_dd8a07a1_acctl across Accounts, Projects and Documents
- E2E_QA_20260513_dd8a07a1_vendor1 in the production vendor picker
- TH/RD CODE FINAL PHASE duplicated in the Pipeline LEAD column
- Four identical PO-0002 rows in Deliveries, all with vendor "--"

Step 1: inventory every QA/E2E-seeded record and report it to me BEFORE deleting anything. I want to see the list.

Step 2: for the four identical PO-0002 rows – determine whether these are genuine duplicate rows or a join fanout in the Deliveries query. Do not delete until you know which. Report your finding.

Step 3: move demo data to a dedicated demo tenant; purge from the ABI tenant.

Do not delete any record referenced by a real record without showing me the reference first.

B-02 • Business days + audit

Execute WO-02 from docs/PRD.md.

Build a business-day service with a Philippine holiday calendar maintained as data, not code (national regular holidays and special non-working days, editable without a deploy). Every SLA in ABI's process deck is in business days; CX ticket clocks are calendar hours – support both.

Then add immutable audit events on all mutable tables: actor, table, row id, before, after, timestamp. Audit rows must be insert-only.

Tests must cover: arithmetic across a year boundary, across Holy Week, and a range that starts on a holiday.

B-03 • Process & SLA engine

Execute WO-03 from docs/PRD.md.

Implement the M-06 tables and seed process_steps from ABI's SD Framework deck (~70 steps, each with responsible BU, input, input-from, output, output-by and an SLA in business days).

Critical distinctions:

- INTERNAL clocks are enforceable and escalate against a BU.
- EXTERNAL clocks (LGU permit, Building Admin vetting, client signature) are tracked but NEVER escalate against a BU. ABI cannot make an LGU move faster.
- Do NOT seed any step whose owner is unresolved. The deck records one step's owner literally as "Commercial or SD - PM?" – leave it out and flag it.

Thresholds: at-risk 80%, breach 100%, escalate 150%.

Ship this in OBSERVE MODE: report breaches at BU level, not individual level. Individual-level enforcement is a later, separate decision.

B-04 · Grain discrimination

Execute WO-04 from docs/PRD.md. This is the first structural migration – go careful

Run migration M-01 on bom_line_items. Backfill the kind discriminator by UOM:

sqm, cu.m, m2, lm, lot	-> work_item
pc, pcs, kg, set, liters	-> material_line
anything else	-> review_queue

Then build the review UI where an estimator confirms classification and EXPLICITLY attaches a material line to a parent work item.

ABSOLUTE RULE: do not auto-reparent anything. Attaching a material line to the wrong work item silently corrupts a cost hierarchy and is nearly undetectable later. Every parent link is a human decision.

Do not enable the I-03 trigger until the review queue is empty.

Before and after the migration, assert that PO, RFQ, cost and budget row counts AND their target ids are byte-identical. Show me that comparison.

Run on staging first. Show me the migration before you run it anywhere.

B-05 · Location dimension

Execute WO-05 from docs/PRD.md.

Run M-02. Parse the leading "<Room> - " prefix out of bom_line_items.description into the project_locations table. Retain the untouched original in description_original.

Example: "Entrance Hall - Vinyl Plank Flooring" becomes item "Vinyl Plank Flooring" at location "Entrance Hall".

Parsing is heuristic. Anything that does not match a clean prefix pattern goes to a review queue – do not mangle it silently.

Acceptance: a query returns total Vinyl Plank Flooring sqm across all locations in a project. Right now that is impossible because the same item exists as three unrelated rows across three rooms.

B-06 • DUPA engine — the critical path

Execute WO-06 from docs/PRD.md. This is the heart of the refactor. Read section 4 the PRD twice before writing anything.

Build the engine BEFORE any UI.

Implement exactly:

```
A Material      = sum(qty x unit_rate)
B Labour        = sum(persons x hourly_rate / productivity_per_hour)
C Equipment     = sum(units x hourly_rate / productivity_per_hour)
D Direct        = A + B + C
E Indirect      = OCM (800 bps of D) + Profit (700 bps of D)
F VAT           = 1200 bps x [base per the vat_base setting]
G Total         = D + E + F
H Unit Rate     = G / header_quantity
```

Rounding: compute A-F unrounded (exact decimal or scaled integers, NEVER float). Round half-up to centavos ONLY when persisting unit_rate_centavos and total_cost_centavos, and when rendering.

D-1: ABI's own sheet labels row F as "12% (D+E)" but arithmetically computes 12% of D – and only that reading makes the sheet foot. Implement vat_base as a configurable enum defaulting to 'direct_only'. Label it honestly in the UI.

Write these two tests first, before the implementation:

TEST 1 (canonical, from ABI's MNHPI workbook):

```
Item: "Stair footing concrete inc. forms", unit cu.m, header_quantity 0.10
Material: 1.00 cu.m @ P5,951.00
Labour:   Foreman 1.00 @ P272.02 productivity 1.00
          Skilled 1.00 @ P199.09 productivity 1.00
```

Non-Skilled 2.00 @ P173.79 productivity 1.00
Equipment: One-Bagger Concrete Mixer, 1 unit @ P600.00/hr, productivity 0.10
EXPECT: G = 1621750 centavos, H = 16217500 centavos, exactly.

TEST 2 (D-2 regression):

A BOQ unit cost sourced from G instead of H must fail the build. In ABI's shipped MNHPI BOQ this exact error understated one line by 10x – the BOQ reads unit cost P16,217.50 where H is P162,175.00. This test exists so that class of error becomes structurally impossible.

Then build the assembly, crew, equipment, material-catalog and price-history libraries. Changing a crew rate must recompute every DUPA that references it.

B-07 · BOM Builder rebuild

Execute WO-07 from docs/PRD.md.

Rebuild the BOM Builder as a VIEW over the new model. Do not touch the underlying bom_line_items ids.

New structure: BOQ divisions with sub-totals -> work items with DERIVED unit rates -> DUPA detail behind progressive disclosure.

Keep (these instincts are correct): version numbering, "Submit for Client Approval", the supplier switcher panel, the pricing-breakdown chips.

Remove: the editable UNIT COST field wherever unit_rate_source='dupa', and the line-level MARKUP % input. The current form defaults MARKUP % to 30 and the whole BOM prices at exactly x1.30 – that is what we are eliminating.

Fix: the supplier switcher currently reports "No rate cards match this line" on every AI-drafted row because it joins on description text. Join on catalog_item_id instead.

Fix: the pricing-breakdown chips currently read RAG 0 / Catalog 0 / Manual 0 / Unpriced 0 against 13 lines. They must report real state.

B-08 · Takeoff importer

Execute WO-08 from docs/PRD.md.

Build a GENERIC structured-takeoff importer. Togal is one producer among several, never a dependency.

Flow: file -> preview -> saved column-mapping profile per source -> validation (UOM recognised? division assigned? duplicate detected?) -> UPSERT into bom_line_items keyed on (takeoff_import_id, source_row_key).

Bind every imported line to a drawing_revision_id.

THE BLOCKING TEST (I-10): assign a vendor to an imported line, re-run the same import, and the vendor assignment must survive. Delete-and-reinsert on re-import would orphan every downstream PO – if your implementation does that, it is wrong.

Manual entry must remain available as a fallback at all times.

B-08a · Existing AI CAD auto-draft disposition

Execute WO-08a from PRD.md, immediately after WO-08.

The build already ships "Auto-drafted from CAD upload". It stays. Do not delete it and do not switch it off.

What changes: its output stops being a priced BOM line and becomes one more producer feeding the same pipeline as any other takeoff source.

1. Route AI output through the IDENTICAL validation gate as WO-08. No privileged path because the model was confident.
2. Land AI rows as kind='work_item' in draft state, with NO unit rate and unit_rate_source='manual'. An extracted description is scope candidate, not a price.
3. Every row failing validation goes to an UNRESOLVED QUEUE with a stated reason: no UOM match / no division / no catalog match / quantity defaulted. Nothing is silently accepted.
4. Retain the AI provenance tag plus source model, drawing revision and extraction timestamp. This is the evaluation corpus for the post-MVP mapping layer – do not discard it.
5. HARD GATE: a BOM cannot be submitted for client approval while its Unresolved queue is non-empty.

Do NOT build the AI mapping layer here. Description normalisation, UOM inference, automatic division assignment and assembly matching are post-MVP.

The current failure mode you are fixing: a line sits at qty 1, vendor unassigned, "No rate cards match this line", and looks finished. It is not finished. Make that visible.

Acceptance: re-running extraction over the same drawing revision upserts, does not duplicate, does not clear vendor assignments, and never overwrites a line that already has a DUPA.

B-09 · Excel importers

Execute WO-09 from docs/PRD.md.

Build importers for ABI's real Excel templates: PPRF, SI Report, BOQ/BOM, BOE, Level 1 Master Schedule. I am attaching the real files.

This is the adoption strategy, not a migration chore. ABI's entire operating history lives in files named like Project_PPRF_Ver3_11082024.xlsx and Project_BOE_Ver2_11062024.xlsx. A system they cannot load these into will be worked around rather than worked in.

Importing historical BOEs must seed the assembly, crew and material libraries – that is what makes the DUPA engine fast on day one.

Every importer needs: preview, column mapping, validation, and a rejected-rows report. Nothing imports silently. Every skipped row is shown to the user with a reason.

B-10 · RFQ to price history

Execute WO-10 from docs/PRD.md.

Capture supplier quotes as structured lines rather than attachments. Build the price comparison matrix. Write quoted and awarded prices to price_history with vendor and date.

Surface suggestions inside the DUPA material line, always showing source and date. Any rate older than 90 days renders with a staleness warning.

This closes the loop that currently runs through email: Procurement issues RFQs to support Commercial's BOM generation, quotes come back as attachments, and prices get retyped into the BOE.

B-11 · PPRF + KYC gate

Execute WO-11 from docs/PRD.md.

Build the structured PPRF form. Submitting it creates Client + Opportunity in one action – this replaces the versioned PPRF spreadsheet.

Then the KYC gate, which is TWO independent tracks with separate 2-day clocks:

- Financial Evaluation Report: Finance-GA produces, FC recommends, President approves
- Credit Investigation Report: Finance-AR produces, FC notes, President endorses

Inputs: AFS current year + previous 2 years, BIR Certificate 2303, VAT exempt/zero-rated certification, KYC including top 10 suppliers and top 10 clients.

Keep the existing KYC Queue UI – the gating concept there is already right. An opportunity cannot advance past Site Survey until both tracks clear.

B-13 • Award automation

Execute WO-13 from docs/PRD.md. This is the demo moment – build it to be shown.

Trigger: signed BOM uploaded.

In ONE atomic transaction, create:

- Project (with Project Code)
- Budget baseline from the approved BOQ
- Cost codes from the BOQ divisions
- AR Code request to Finance-AR (2 day clock)
- DP invoice draft (2 day clock)
- CARI / bond task (5-10 day clock)
- Project Tracker (2 day clock from NTP)
- CX onboarding task (7 days after NTP)

Zero fields retyped between the signed BOM and the live project. The transaction must be atomic and reversible.

Today this hand-off is where ABI loses the most time and accuracy – the project identity gets created three separate times across PPRF, BOM and AR/Project code.

B-15 • RFPO to PO to commitment

Execute WO-15 from docs/PRD.md.

DO NOT rewrite the existing PO module. Its approval timeline (Draft -> PM approved -> Commercial approved -> SCM issued -> Supplier notified) and its PH tax handling (12% VAT, 2% withholding) are the best-built parts of the system. Carry them forward unchanged.

Add on top:

- The RFPO stage (Commercial endorses to Procurement with approved finishes and specs)
- Delegation-of-Approval routing by amount band [PASTE ABI MATRIX HERE]
- Commitment against budget_lines, joined on bom_line_item_id

Issuing a PO must reduce remaining allowable on the cost code. A PO that would exceed allowable warns before submission.

B-16 · Permits and mobilization gate

Execute WO-16 from docs/PRD.md.

Permit types: LGU building permit, Building Admin vetting (15 d), DOLE, Occupancy Permit, CARI, and performance/surety/construction bonds tracked through to REFUND.

Each carries: responsible person, submitted date, expected return, actual return, live countdown, escalation.

LGU duration is contested in ABI's own documents – the process deck says 45-60 days, a live project schedule shows 30. Model it as min/expected/max PER LGU and learn from actuals. Do not hardcode.

The headline feature is the MOBILIZATION READINESS GATE: a single tile showing the four external returns as done/waiting with days-at-risk –

1. Commented FCD from Building Admin
2. PO copies from Procurement
3. CARI from Finance
4. NTP from Building Admin

Mobilization cannot be marked started until all four land, or an authorised override is logged with a reason and an actor.

B-17 · Cost control

Execute WO-17 from docs/PRD.md.

Build: Budget -> Committed (POs) -> Actual (PO-invoiced) -> Remaining -> Variance -> Live margin. Drilldown by cost code. Every join is on bom_line_item_id.

Acceptance for this one is a HUMAN sign-off, not a test: Commercial must confirm the live margin on a running project matches their own spreadsheet. Build it so that comparison is easy to make – show the workings, not just the number.

B-18 · Management dashboard

Execute WO-18 from docs/PRD.md.

Keep the existing metric selection – Active Pipeline TCV, Active GP, blended margin, Weighted Pipeline, Closed Won FYTD are the right metrics at the right altitude.

Add: margin by project with variance, cost variance against budget, permit exposure, unsigned-VO exposure, and SLA breaches by BU.

Target: the President runs the Monday meeting from this screen without opening Excel.

C · Control prompts

C-1 · Drift reset — use the moment a session starts wandering

Stop. You are drifting from the PRD.

Re-read the constraints in AGENTS.md and docs/PRD.md and confirm each one back to

1. This is a refactor. bom_line_items.id is a stable UUID already referenced by cost, budget, PO and RFQ. It stays.
2. No scope_items table. No scope_item_id column. Ever.
3. Additive migrations only. No DROP. No foreign key re-pointed.
4. No general ledger work.
5. No float in any monetary path.

Then discard whatever you just produced and restart the current work order from the text in docs/PRD.md. Do not attempt to repair drifted output – repaired drift to review than fresh work.

C-2 · Migration review — run before any migration touches staging

Before I run this migration, review it against the PRD and answer each point explicitly:

1. Is it purely additive? List every ALTER and CREATE.
2. Does it DROP anything, or re-point any existing foreign key?
3. Does every new table have tenant_id NOT NULL and a matching RLS policy?
4. Does any new monetary column use anything other than BIGINT centavos?
5. What breaks if this runs twice? Is it idempotent?

6. What is the exact rollback? Write it out.
7. Which downstream tables reference the rows this touches, and how do I prove those references are unchanged after it runs?

Answer all seven. Do not summarise.

C-3 · Work-order sign-off

Work order [WO-NN] is claimed complete. Verify it against the PRD:

1. List each acceptance criterion from the PRD for this work order.
2. For each, name the test that proves it and show me the test.
3. Any criterion without an automated test is NOT met – say so plainly.
4. Which of the §6 invariants does this work order touch, and are they still enforced?
5. What did you change that the PRD did not ask for? Justify each item.

Be adversarial. I would rather find a gap now than after the next work order builds on it.

C-4 · Estimator design review — for WO-06 and WO-07

I am about to show this to an ABI estimator who currently builds ~55 Detailed Unit Price Analysis sheets per project by hand in Excel.

Walk me through the exact click path for building one DUPA for a suspended ceiling, mineral fibre 600x600, 120 sqm, using the assembly library.

Count the keystrokes and the clicks. Compare honestly against doing it in Excel.

If it is not faster than Excel, tell me that plainly and tell me where the time is going. The MVP does not launch unless this is faster.

C-5 · Ask before assuming

Before you implement this, list every assumption you are making that is not explicitly stated in the PRD. For each one, tell me whether getting it wrong is cheap or expensive to reverse.

Ask me about anything expensive. Proceed on anything cheap and note it.

D · The five questions to send ABI today

These block work orders. They take days to come back, so send them before you start coding.

To Commercial:

1. In our Detailed Unit Price Analysis sheets, row F is labelled "12% (D+E)" but the arithmetic computes 12% of D only. Which is intended? It changes every unit rate by about 1.4%. (*Blocks WO-06.*)
2. Are the 8% overhead/contingency and 7% profit figures ABI policy, DPWH convention, or specific to that client? (*Affects defaults.*)
3. Who owns keeping crew rates and material prices current, as a named standing responsibility? (*Blocks WO-06 sign-off.*)

To Management / Finance:

4. We need the current ABI Delegation of Approval matrix — amount bands, approver roles, sequence, and escalation. (*Blocks WO-15.*)

To whoever holds the templates:

5. We need live copies of: PPRF, SI Report, BOE, BOQ, Milestone Definition, Project Tracker, Allowable Budget Form, Interim Payment Certificate, Level 1 Master Schedule — and one real Togal export. (*Blocks WO-09 and the Togal mapping.*)

E · Suggested sequencing over calendar time

Weeks	Work orders	Why grouped
0	Pre-flight (§0.3), send §D questions	Answers take days; start the clock
1	WO-00, WO-01, WO-02	Foundation and credibility. Fast wins
2–3	WO-03	SLA engine — everything later inherits it
3–4	WO-04, WO-05	The two structural migrations. Staging first
4–7	WO-06	The critical path. Do not compress this
6–8	WO-09 (<i>parallel with WO-06</i>)	Seeds the libraries WO-06 needs

7–9	WO-07, WO-08	UI and intake over the new model
9–10	WO-10, WO-11, WO-12	Commercial loop closes
10–11	WO-13	Demo to management here
11–13	WO-14, WO-15, WO-16	Budget, procurement, permits
13–15	WO-17, WO-18	Cost truth and the dashboard
15	MVP success test (§12.12)	Estimator faster than Excel, or do not launch

Roughly 15 weeks to MVP with a small team. WO-06 is the item most likely to slip and the item least worth rushing.

F • The one test that decides everything

Everything in the PRD and this pack exists to make one sentence true on launch day:

An ABI estimator produces a complete, client-presentable BOQ and BOE faster in BUILD OPS than in Excel.

If that is false, do not launch. Fix the DUPA engine and the Excel importers, and test again.